Real-Time Innovative Codathon Questions

Level 0 (Easy - 5 Questions)

Focus: Conditional Logic, Basic Calculations, and Real-world Formulas.

Q1: Smart Water Meter (Leak Detector)

Problem: A smart meter monitors water flow per minute. To detect a leak, it checks if a constant flow exists for a duration of 5 minutes without any "zero-flow" intervals.

- **Rules:**
  - If the average flow across 5 measurements is > 2.0L/min AND no single measurement is 0, output `LEAK DETECTED`.
  - Otherwise, output `NORMAL`. **Input:** 5 floats representing flow rate per minute. **Output:** `LEAK DETECTED` or `NORMAL`

Q2: Aviation Luggage Scanner

Problem: A self-service kiosk calculates baggage fees.

- **Base Allowance:** 15 kg.
- **Overweight Fee:** Rs. 500 per kg forทุก kg above 15.
- **Oversize Surcharge:** If `Length + Width + Height > 158cm`, add a flat **Rs. 1000** surcharge. **Input:** Weight (float), Length, Width, Height (ints). **Output:** `Total Fee: Rs. [Value]`

Q3: Cinema Dynamic Pricing

Problem: A theater uses dynamic pricing.

- **Base Price:** Rs. 200.
- **Age Discount:** Age < 12 (50% off), Age > 60 (30% off).
- **Time Surcharge:** If the show is after 18:00 (6 PM), add **Rs. 50** to the final price. **Input:** Age (int), Show Hour (int, 0-23). **Output:** `Ticket Price: Rs. [Value]`

Q4: Fitness Workout Tracker

Problem: Calculate calories burned.

- **Formula:** `(Duration * Intensity) / 10`.
- **Intensity Levels:**
  - Walking: 3, Running: 8, Cycling: 6.
- **Heart Rate Bonus:** If `Avg Heart Rate > 150`, add **10% extra** to the total calories. **Input:** Duration in mins (int), Exercise Type (W/R/C), Avg Heart Rate (int). **Output:** `Calories: [Value] kcal` (round to 1 decimal).

Q5: Smart Vending Machine

Problem: Dispense change using the fewest coins. Available coins: Rs. 10, 5, 2, 1. **Input:** Item Price (int), Amount Paid (int). **Output:** `Change: 10x[a], 5x[b], 2x[c], 1x[d]`

Level 1 (Medium - 5 Questions)

Focus: Data Structures, Windows, and Real-time Processing.

Q1: Stock Market "Golden Cross"

Problem: Detect a Golden Cross, which occurs when a 3-day Moving Average (MA3) crosses above a 5-day Moving Average (MA5). **Input:** 7 days of closing prices (floats). **Output:** Output `GOLDEN CROSS at Day [i]` for the first day where MA3 > MA5 (if it was <= before).

Q2: Flash Sale Inventory Manager

Problem: You have 50 units of a "SuperPhone". Process a list of orders.

- If an order quantity is available, subtract from stock and print `ORDER_FILLED`.
- If partial stock is available, fill the remaining and print `PARTIAL_FILLED: [count]`.
- If zero stock, print `OUT_OF_STOCK`. **Input:** Number of orders (int), then a list of quantities. **Output:** Status for each order.

Q3: Smart Greenhouse Controller

Problem: A greenhouse has 4 sensors. It activates a fan if the average humidity is above 70% OR if any single sensor exceeds 85%. **Input:** 4 humidity percentages (floats). **Output:** `FAN_ON` or `FAN_OFF`.

Q4: Library Late Fee System

Problem: Calculate fines.

- **Rule:** Rs. 2/day for first 5 days. Rs. 5/day after that.
- **Book Type:** `Scientific` books have double the total fine.
- **Holiday Exemption:** If the return was on a Sunday, reduce total fine by Rs. 5. **Input:** Days Overdue (int), Book Type (S for Scientific, O for Other), Is Sunday (1 for Yes, 0 for No). **Output:** `Fine: Rs. [Value]`

Q5: Courier Zone Router

Problem: Route a package based on distance from Hub (0,0).

- **Zone A:** Distance <= 10.0km. Tax: Rs. 50.
- **Zone B:** Distance <= 50.0km. Tax: Rs. 150.
- **Zone C:** Distance > 50.0km. Tax: Rs. 500. **Input:** Customer X, Customer Y (ints). **Output:** `Zone: [A/B/C], Tax: Rs. [Value]`

Level 2 (Hard - 5 Questions)

Focus: Optimization, Complex State, and Signal Processing.

Q1: Energy Load Balancer

Problem: Manage power for a small town.

- **Sources:** Solar, Wind, Grid.
- **Priority:** Use Solar first, then Wind, then Grid.
- **Constraint:** Grid can only supply up to 500 units. If total demand exceeds all sources, print `BLACKOUT`. **Input:** Demand (int), Solar Available (int), Wind Available (int). **Output:** `Usage: S:[a], W:[b], G:[c]` or `BLACKOUT`.

Q2: Cybersecurity Log Parser (Brute Force)

Problem: Detect a Brute Force attack. An attack is defined as **more than 3 failed attempts** for the *same* username within any 10-minute window in the log. **Input:** List of 10 entries: `[TimeInMins] [Username] [Status: S/F]`. (e.g., `5 user1 F`) **Output:** `ATTACK DETECTED: [Username]` or `SYSTEM_SECURE`.

Q3: Factory Assembly Bottleneck

Problem: An assembly line has 5 stations. You are given the processing time (in seconds) for 10 items at each station. **Task:** Identify the station with the highest **average** processing time. This is the bottleneck. **Input:** 5 lines, each containing 10 integers (time per item). **Output:** `Bottleneck: Station [1-5]`

Q4: Healthcare ECG Monitor

Problem: Detect an "Arrhythmia" pattern. An arrhythmia is signaled if the time interval between 4 consecutive heartbeats (R-R interval) varies by more than 20% from the first interval in that set. **Input:** 10 timestamps of heartbeats (ms). **Output:** `Status: NORMAL` or `Status: ARRHYTHMIA_DETECTED`.

Q5: Logistics Multi-stop Optimizer

Problem: A driver visits 4 cities in a specified order.

- **Cost:** Fuel is Rs. 100 per unit distance.
- **Tolls:** Each city has a toll fee.
- **Optimization:** If the total distance is > 500km, the company provides a **Rs. 1000 fuel subsidy**. **Input:** Coordinates of 4 cities (x1 y1, x2 y2, x3 y3, x4 y4) and their 4 toll fees. **Output:** `Total Cost: Rs. [Value]` (Distance formula: $\sqrt{(x2-x1)^2 + (y2-y1)^2}$)